

Course Title: Data Structures and Algorithms

Course Code: CSE 201

Credits: 4(3|1)

Prerequisites: CSE 104

Co-requisites: None

School/Department: School of Computing and Information Technology / Department of Software Engineering

Course Description

The course is a deep dive into the fundamental principles and practices focusing on data structures and algorithms, helping students to model practical problems, and equipping them with the knowledge and tools to construct efficient algorithms for these problems.

The course provides students with knowledge about data structures such as stack, queue, list, hashtable, etc., and algorithms for searching and sorting in lists, with an emphasis on the efficiency of these algorithms and how to use libraries and programming tools effectively. Students practice designing, analyzing, and implementing data structures using a programming language.

Course Objectives

After finishing the course, students must achieve the following goals:

- Evaluate the complexity of different algorithms.
- Design time and memory-efficient programs for practical problems.
- Effectively use libraries and programming tools to enhance performance and code quality.

Course Intended Learning Outcomes

At the end of this course, students will be able to:

CLO1. Implement various data structures (e.g., stack, queue, list, hashtable) and algorithms (e.g., searching, sorting) efficiently.

CLO2. Design and develop data models to accurately represent and solve practical problems.

CLO3. Create time and space-efficient algorithms to solve complex problems.

CLO4. Utilize libraries to enhance coding productivity, efficiency, and quality.

Course Content**Topic 1: Introduction**

1. Introduction to Data Types and Data Structures.

2. Introduction to Algorithms.
3. The role of the course in Software Development.

Topic 2: Programming Techniques

1. Classes and Objects.
2. Data Types, Value ranges, Wrapper classes
3. Data Types and Memory Management
4. Functional Programming Essentials
5. Utility classes
6. Performance Enhancement Techniques
7. Practical Problems

Topic 3: Algorithms

1. Algorithm Definition
2. Algorithm Complexity and Runtime Analysis
3. Comparative Analysis
4. Recursion
5. Two Pointers
6. Sliding Window
7. Prefix Sum

Topic 4: Abstract Data Types and Applications

1. Comprehensive Overview of Data Structures
2. Java Built-in Data Types and Data Structures
3. Stack, Queue and practical problems
4. Dictionary and practical problems

Topic 5: Custom Data Structure Design

1. Designing Collections
2. Generalization
3. Iterator
4. Custom ListNode, LinkedList

Topic 6: Practical Data Modeling

1. User-Defined Data Types
2. Properties and Action Methods
3. Implementing Comparable
4. Problem Solving Paradigms
5. Real-World Scenarios

Topic 7: Sorting

1. Basic Sorting Algorithms

2. Advanced Sorting Techniques
3. Modern Sorting in Practice

Topic 8: Efficient Searching Techniques

1. Linear Search
2. Binary Search (Item and Values)
3. Built-in Binary Search
4. Interpolation Search
5. Application of Searching Algorithms

Topic 9: Problem Solving with Data Types and Algorithms

1. Advanced Binary Search Techniques
2. Designing Complex Data Types
3. Priority Queue and Practical Problems
4. Algorithm Optimization Strategies

Course Materials¹ & Other Requirements

Textbook:

Frank M. Carrano. (2019). *Data Structures and Abstractions with Java (5th ed.)*.
Prentice Hall.

Janet J. Prichard, Frank M. Carrano. (2011). *Data Abstractions & Problem Solving with Java (3rd ed.)*. Addison-Wesley

Required Readings: None

Additional Readings: None

Other requirements (software, equipment, online resources, etc.):

<http://visualgo.net/>

<http://docs.oracle.com/javase/8/docs/api/>

Course Assessment ²

Assessment Type	Assessment Method	Description	Weight (%)		Aligned CLO
Formative	[1] Homework (HW)	Solve a wide range of practical problems throughout the course.	40	25	1 – 4

¹ Available at the EIU Library

² The sum of x, y, and z must be 100%;
 $\Sigma x_i = \Sigma y_i = \Sigma z_i = 100\%$

	[2] Practical test in class	Solve practical problems using built in data structures and well-known algorithms.	25	1 – 4
	[3] Practical test in class	Solve practical problems in specific domains that require design appropriate data structures.	25	1 – 4
	[4] AI-assisted group project	Students work in teams of 2–3 to solve a complex problem using algorithmic design and optimization strategies with support from AI tools. Each team must develop an optimal solution, document their process, and present their results to the lecturer.	25	1 – 4
Final	[5] Exam	Solve practical problems ranging from easy to hard difficulty.	60	1 – 4

Course Schedule³

Week 1 Lesson 1 2 hrs. theory Aligned CLO: 1, 2	Topic 1: Introduction Teaching methods: Lecture Reading - Chapter 1, 2, 3 - <i>Data Abstractions & Problem Solving with Java</i> (Janet J. Prichard et al., 2011, pp. 1–27, 55-78, 91-117) - Chapter Introduction Organizing Data, Prelude, 3 - <i>Data Structures and Abstractions with Java</i> (Frank M. Carrano et al., 2019, pp. 48–50, 19-44, 139-176) Discussion - Importance of data structures - Algorithm efficiency - Software development challenges Assessment: None
Week 1 Lesson 2	Topic 2: Programming Techniques Teaching methods: Lecture

³ Students need two hours of self-study for each hour of classroom instruction.

2 hrs theory Aligned CLO: 1, 2, 3, 4	Reading - Chapter 2 – <i>Data Abstraction & Problem Solving with Java</i> (Prichard et al., 2011, pp. 55–78) - Chapter Prelude, Appendix B – Data Structures and Abstractions with Java (Carrano et al., 2019, pp. 19–44, 48–50) Discussion - Functional Programming Essentials - Data Types - Wrapper classes Assessment: None
Week 2 Lesson 1 2 hrs theory 2 hrs practice Aligned CLO: 1, 2, 3, 4	Topic 2: Programming Techniques (cont.) Teaching methods: Lecture Reading - Chapter 2 – <i>Data Abstraction & Problem Solving with Java</i> (Prichard et al., 2011, pp. 55–78) - Chapter Prelude, Appendix B – Data Structures and Abstractions with Java (Carrano et al., 2019, pp. 19–44, 48–50) Discussion - Classes and Objects - Utility classes Practice - Lab 1 – Practice problems: https://www.spoj.com/EIUPROGR/ Assessment: [Homework]
Week 2 Lesson 2 2 hrs theory Aligned CLO: 1, 2, 3, 4	Topic 3: Algorithm Teaching methods: Lecture Reading - Chapter 3 – <i>Data Structures and Abstractions with Java</i> (Carrano et al., 2019, pp. 105–168) Discussion - Algorithm Definition - Complexity Analysis - Recursion Assessment: [Homework]
Week 3	Topic 4: Abstract Data Types and Applications

Lesson 1 2 hrs theory 4 hrs practice Aligned CLO: 1, 2, 3, 4	Teaching methods: Lecture Reading - Chapter 5, 6 – <i>Data Abstraction & Problem Solving with Java</i> (Prichard et al., 2011, pp. 139–176) - Chapter 5, 6 – <i>Data Structures and Abstractions with Java</i> (Carrano et al., 2019, pp. 139–176) Discussion - Classification of data structures - Java built-in collections - Stack and queue applications - Dictionary applications Practice - Lab 2 – Practice problems: https://www.spoj.com/EIUPROGR/ Assessment: [Homework]
Week 3 Lesson 2 2 hrs theory 2 hrs practice Aligned CLO: 1, 2, 3, 4	Topic 4: Abstract Data Types and Applications (cont) Teaching methods: Lecture Reading - Chapter 5, 6 – <i>Data Abstraction & Problem Solving with Java</i> (Prichard et al., 2011, pp. 139–176) - Chapter 5, 6 – <i>Data Structures and Abstractions with Java</i> (Carrano et al., 2019, pp. 139–176) Discussion - Classification of data structures - Java built-in collections - Stack and queue applications - Dictionary applications Practice - Lab 3 – Practice problems: https://www.spoj.com/EIUPROGR/ Assessment: [Homework]
Week 4 Lesson 1 2 hrs theory 2 hrs practice	Topic 5: Custom Data Structure Design Teaching methods: Lecture Reading Chapter 8, 9 – <i>Data Abstraction & Problem Solving with Java</i> (Prichard et al., 2011, pp. 245–296)

Aligned CLO: 1, 2, 3, 4	- Chapter 10–13 – <i>Data Structures and Abstractions with Java</i> (Carrano et al., 2019, pp. 300–365) Discussion - Designing collections for specific applications - Generalization of data structures - Iterator as an access mechanism - Linked node and linked list customization Practice - Lab 3 – Practice problems: https://www.spoj.com/EIUPROGR/ Assessment: [Homework]
Week 4 Lesson 2 2 hrs theory 4 hrs practice Aligned CLO: 1, 2, 3, 4	Topic 5: Custom Data Structure Design (cont) Teaching methods: Lecture Reading - Chapter 8, 9 – <i>Data Abstraction & Problem Solving with Java</i> (Prichard et al., 2011, pp. 245–296) - Chapter 10–13 – <i>Data Structures and Abstractions with Java</i> (Carrano et al., 2019, pp. 300–365) Discussion - Designing collections for specific applications - Generalization of data structures - Iterator as an access mechanism - Linked node and linked list customization Practice - Lab 4 – Practice problems: https://www.spoj.com/EIUPROGR/ Assessment: [Homework]
Week 5 Lesson 1 2 hrs theory Aligned CLO: 1, 2, 3, 4	Topic 6: Practical Data Modeling Teaching methods: Lecture Reading - Chapter 7 – <i>Data Abstraction & Problem Solving with Java</i> (Prichard et al., 2011, pp. 210–244) - Chapter 9, 14 – <i>Data Structures and Abstractions with Java</i> (Carrano et al., 2019, pp. 280–310, 400–420) Discussion - User-defined data types and applications

	Practice Lab 4 – Practice problems: https://www.spoj.com/EIUPROGR/ Assessment: [Homework, Practice test in class 1]
Week 5 Lesson 2 2 hrs theory 2 hrs practice Aligned CLO: 1, 2, 3, 4	Topic 6: Practical Data Modeling (cont) Teaching methods: Lecture Reading - Chapter 7 – <i>Data Abstraction & Problem Solving with Java</i> (Prichard et al., 2011, pp. 210–244) - Chapter 9, 14 – <i>Data Structures and Abstractions with Java</i> (Carrano et al., 2019, pp. 280–310, 400–420) Discussion - User-defined data types and applications Practice - Lab 5 – Practice problems: https://www.spoj.com/EIUPROGR/ Assessment: [Homework]
Week 6 Lesson 1 2 hrs theory 2 hrs practice Aligned CLO: 1, 2, 3, 4	Topic 7: Sorting Teaching methods: Lecture Reading - Chapter 11 – <i>Data Abstraction & Problem Solving with Java</i> (Prichard et al., 2011, pp. 320–360) - Chapter 15, 16 – <i>Data Structures and Abstractions with Java</i> (Carrano et al., 2019, pp. 430–490) Discussion - Sorting algorithms Practice - Lab 6 – Practice problems: https://www.spoj.com/EIUPROGR/ Assessment: [Homework]
Week 6 Lesson 2 2 hrs theory 2 hrs practice Aligned CLO: 1, 2, 3, 4	Topic 7: Sorting (cont) Teaching methods: Lecture Reading - Chapter 11 – <i>Data Abstraction & Problem Solving with Java</i> (Prichard et al., 2011, pp. 320–360) - Chapter 15, 16 – <i>Data Structures and Abstractions with Java</i> (Carrano et al., 2019, pp. 430–490)

	Discussion • Sorting algorithms Practice • Lab 6 – Practice problems: https://www.spoj.com/EIUPROGR/ Assessment: [Homework, Group Project]
Week 7 Lesson 1 2 hrs theory 2 hrs practice Aligned CLO: 1, 2, 3, 4	Topic 8: Efficient Searching Techniques Teaching methods: Lecture Reading • Chapter 12 – <i>Data Abstraction & Problem Solving with Java</i> (Prichard et al., 2011, pp. 361–400) • Chapter 19 – <i>Data Structures and Abstractions with Java</i> (Carrano et al., 2019, pp. 520–560) Discussion • Searching algorithms Practice • Lab 6 – Practice problems: https://www.spoj.com/EIUPROGR/ Assessment: [Homework]
Week 7 4 hrs theory 4 hrs practice Aligned CLO: 1, 2, 3, 4	Topic 9: Problem Solving with Data Types and Algorithms Teaching methods: Lecture Reading • Chapter 13 – <i>Data Abstraction & Problem Solving with Java</i> (Prichard et al., 2011, pp. 401–450) • Chapter 24–26 – <i>Data Structures and Abstractions with Java</i> (Carrano et al., 2019, pp. 600–670) Discussion • Advanced Binary Search Techniques • Designing Complex Data Types • Priority Queue Techniques • Algorithm optimization Practice • Lab 7 – Practice problems: https://www.spoj.com/EIUPROGR/ Assessment: [Homework, Practice test in class 2]
Week 8	Topic 9: Problem Solving with Data Types and Algorithms (cont) Teaching methods: Lecture

2 hrs theory 2 hrs practice Aligned CLO: 1, 2, 3, 4	Reading - Chapter 13 – <i>Data Abstraction & Problem Solving with Java</i> (Prichard et al., 2011, pp. 401–500) - Chapter 24–26 – <i>Data Structures and Abstractions with Java</i> (Carrano et al., 2019, pp. 600–720) Discussion - Advanced Binary Search Techniques - Designing Complex Data Types - Priority Queue Techniques - Algorithm optimization Practice - Lab 7 – Practice problems: https://www.spoj.com/EIUPROGR/ Assessment: [Homework]
Week 9 4 hrs practice Aligned CLO: 1, 2, 3, 4	Reading: None Discussion: None Practice Lab 8 – Practice problems: https://www.spoj.com/EIUPROGR/ Assessment: [Homework]
Week 10 2 hrs practice Aligned CLO: 1, 2, 3, 4	Reading: None Discussion: None Practice Lab 9 – Practice problems: https://www.spoj.com/EIUPROGR/ Assessment: [Homework, Practice test in class 3]

Course Policies

- Attendance: If a student is absent 20% or more of the classes, they will not be allowed to take the final examination.
- Similar code is considered cheating; this is a standard policy in software development. Any cheating will result in a score of zero for the assignment, test, or exam.
- Students submit homework after 14 days (score = 0, regardless of accuracy).

Ho Chi Minh City, / /2025

Dean

Program director

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APPENDIX

Appendix 1. Rubrics ⁴

Rubric [1, 2, 3, 5]

CRITERIA		Weight (%)	Excellent	Good	Satisfactory	Failure	Point
			80 - 100	65 – 79.9	50 – 64.9	0 – 49.9	
Solution Quality (Correctness & Algorithmic Design)	CLOs	100	Passes more than 80% of test cases.	Passes more than 65% of test cases.	Passes more than 50% of test cases.	Passes less than 50% of test cases.	

⁴ The rubrics depend on each school's policies, concentrations and courses

Rubric [4]

CRITERIA	CLOs	Weight (%)	Excellent	Good	Satisfactory	Failure	Point
			80 - 100	65 – 79.9	50 – 64.9	0 – 49.9	
Algorithmic Correctness & Optimization	1,2,3,4	50	Solution passes $\geq 80\%$ of test cases and/or ranks within top 20% of best performance on leaderboard; algorithm is efficient and meets time limits.	Passes $\geq 65\%$ of test cases and/or within top 35%; minor inefficiencies.	Passes $\geq 50\%$ of test cases and/or within top 50%; partially optimized.	Passes $< 50\%$ or includes hard-coding; fails to meet performance limits.	
LLM-Guided Problem-Solving Process		20	Clear documentation of LLM use, shows iterative refinement and critical evaluation.	Adequate documentation and reasoning.	Limited use or reflection on LLM support.	Over-reliance on LLM, no documentation.	
Conceptual Understanding & Presentation		30	Team explains logic, complexity, and optimization confidently and accurately.	Mostly clear explanation, minor conceptual gaps.	Basic explanation, limited understanding of design choices.	Unable to explain approach or shows misunderstanding.	

Appendix 2. Mapping Between CLO And PLO

CLO	PLO	PLO
	Software Engineering	CNDC
1. Implement various data structures (e.g., stack, queue, list, hashtable) and algorithms (e.g., searching, sorting) efficiently.	3.1	2.1, 2.2
2. Design and develop data models to accurately represent and solve practical problems.	3.1, 4.1, 4.2	2.1, 2.2
3. Create time and space-efficient algorithms to solve complex problems.	3.2	6.2
4. Utilize libraries to enhance coding productivity, efficiency, and quality.	3.2	2.1, 2.2

Homework Blueprint

Assessment content	CLO	Remember	Understand	Apply	Analyze	Evaluate	Create	Total ques.	Mark
Solve a wide range of practical problems throughout the course. The problems require students to:									
• Design data structures and implement a program to solve practical problems.	1 - 4			1	1	1	1	35-40	100
• Design models and algorithms to solve complex practical problems within constraints of time and memory.									
							Total	35-40	100

Practice Test and Final Examination Blueprint

Assessment content	CLO	Remember	Understand	Apply	Analyze	Evaluate	Create	Total ques.	Mark
I. Implement a program to solve common or well-known programming problems.	1			1				1	30
II. Design data structures and implement a program to solve practical problem.	3				1		1	1	30
III. Design models and algorithms to solve a complex practical problem within constraints of time and memory.	2,3,4				1	1	1	2	40
							Total	4	100